

Chapter 3: Unsupervised Learning

Course material

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In this chapter

- ① **3.1 Clustering:** grouping data without labels
- ② **3.2 Dimensionality reduction:** simplifying and visualizing complex data
- ③ **3.3 Anomaly detection:** identifying rare or suspicious cases

What is unsupervised learning?

- There are no labels: we only have input data X , no correct answers y
- The system must discover the **latent structure** of the data on its own
- Yann LeCun's quote: *"If intelligence is a cake, unsupervised learning is the cake, supervised learning is the icing, and reinforcement learning is the cherry"*

Example: the vast majority of real-world data is unlabeled: photos, text, transactions... that's why this paradigm matters so much in practice.

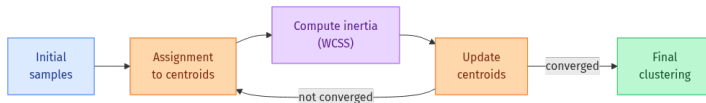
- **Market segmentation:** grouping customers into homogeneous niches without defining the categories in advance
- **Visualization:** projecting data with hundreds of variables into 2D or 3D to understand it at a glance
- **Recommendation systems:** grouping users or products with similar tastes
- **Anomaly detection:** spotting fraud or manufacturing defects never seen before

What is clustering?

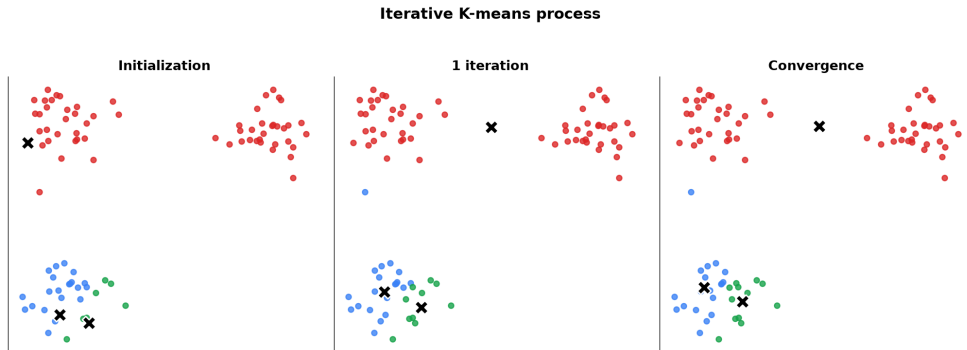
- Goal: split the data into subgroups with **high internal similarity** and **high external differentiation**
- There's no “ground truth”: the system must discover the structure on its own
- Main algorithms: K -means, hierarchical, DBSCAN

K-means: the algorithm's cycle

- 1 **Initialization:** K centroids are chosen
- 2 **Assignment:** each sample joins its nearest centroid
- 3 **Update:** each centroid is recomputed as the mean of its group
- 4 Repeat until the centroids stop moving (**convergence**)



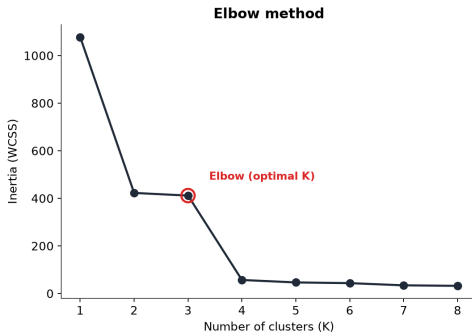
K-means: the process, visually



Example: it's like seating guests at a party: each guest sits at the nearest table, then each table recenters on its guests, and this repeats until no one moves.

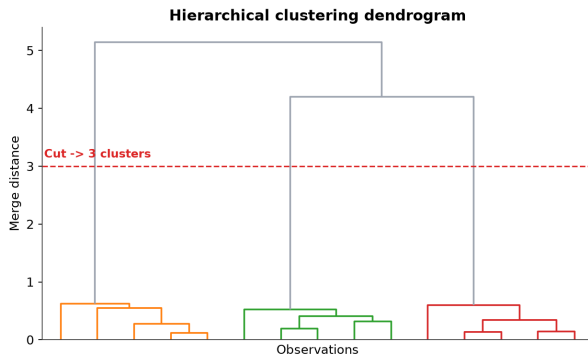
K-means++ and the elbow method

- K-means is sensitive to initialization: poorly placed centroids can lead to a suboptimal result
- K-means++: picks initial centroids as far apart as possible, speeding up convergence
- How many groups K to choose? The **elbow method**: find the point where inertia stops dropping sharply



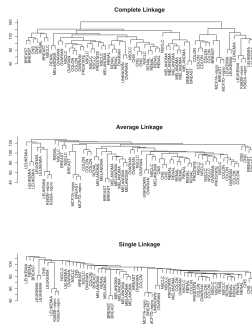
Hierarchical clustering

- Doesn't require fixing K in advance
- **Agglomerative** approach (bottom-up): each point starts as its own group, and at each step the two closest groups merge
- The result is visualized as a **dendrogram**; cutting it at a given height determines the final number of groups



Linkage criteria

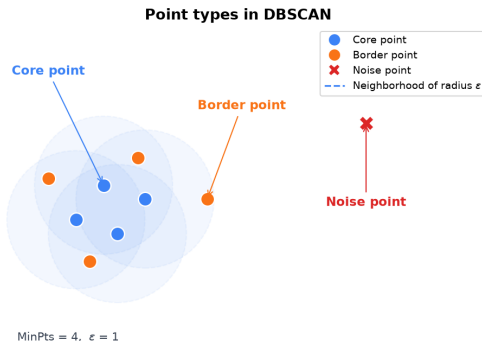
- Define how to measure distance between two groups, not between two points
- **Complete**: *maximum* distance between members — compact, balanced groups
- **Single**: *minimum* distance — sensitive to noise, tends to chain elongated groups
- **Average**: mean of all distances — a balance between the two above



Source: James, Witten, Hastie & Tibshirani (2013), *An Introduction to Statistical Learning*, Fig. 10.17.

DBSCAN: density-based clustering

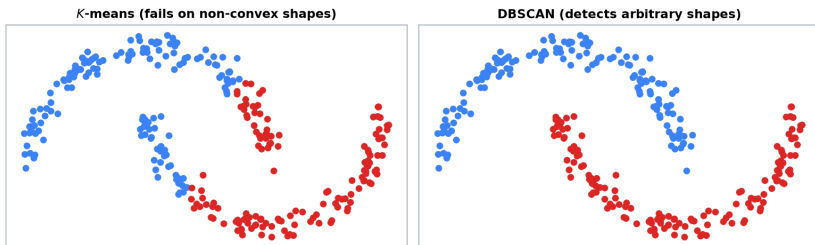
- Defines groups by **local density**, not by closeness to a centroid
- Two parameters: ϵ (neighborhood radius) and **MinPts** (minimum neighbors to be dense)
- Three point types: **core** (dense), **border** (near a core point), and **noise** (isolated)



DBSCAN vs. K -means

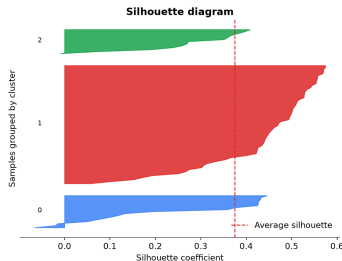
- DBSCAN detects **arbitrary shapes** (not just spherical) and isolates noise natively
- K -means forces every point into a group, even outliers

K -means vs. DBSCAN on moon-shaped data



Evaluation: silhouette coefficient

- Without labels, how do we know if a clustering is good?
- The silhouette coefficient compares, for each point, its **cohesion** (distance to its own group) with its **separation** (distance to the nearest other group)
- Range $[-1, 1]$: near $+1$ is a good grouping; near 0 , an ambiguous boundary; negative, likely misassigned
- Other criteria exist (AIC/BIC, purity, Rand index...) to compare against already-known labels



Summary: 3.1 Clustering

- **K-means**: simple and fast, but needs a known K and roughly spherical groups
- **Hierarchical**: no K needed, gives a visual, interpretable dendrogram
- **DBSCAN**: detects arbitrary shapes and isolates noise automatically
- **Silhouette**: standard metric to evaluate quality without labels

The curse of dimensionality

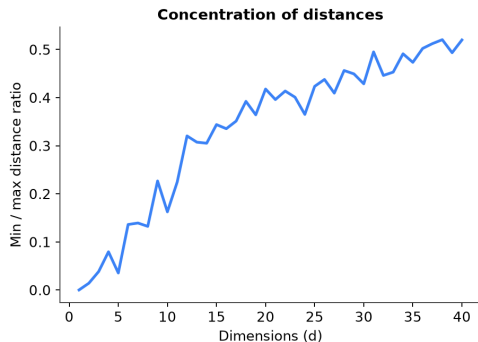
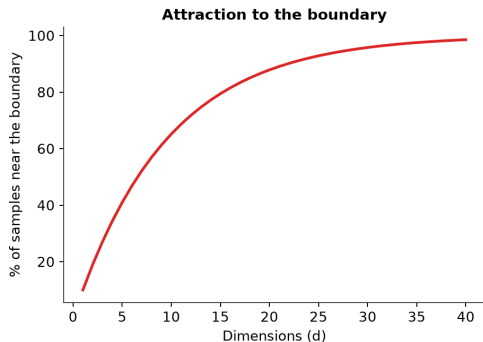
- As dimensions increase, the space becomes **ultra-sparse**
- Samples tend to sit at the boundary of the space, not in the center
- Distances between points become nearly indistinguishable, degrading distance-based algorithms (like K -NN)



Why reduce dimensions?

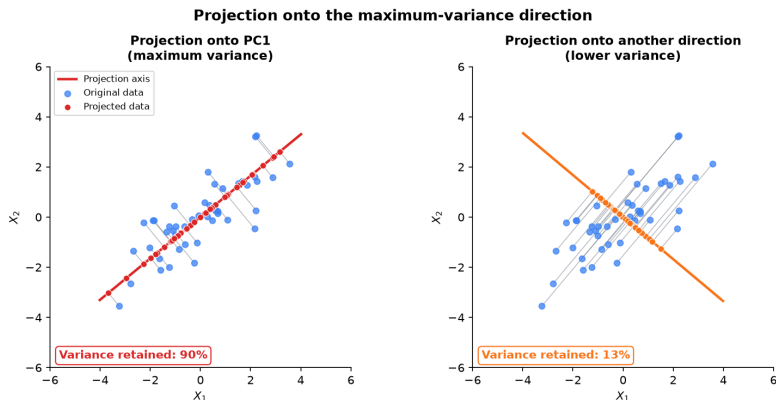
- Drastically speeds up computation
- Mitigates overfitting
- Indispensable for **visualizing** complex data in 2D or 3D

The curse of dimensionality, quantified



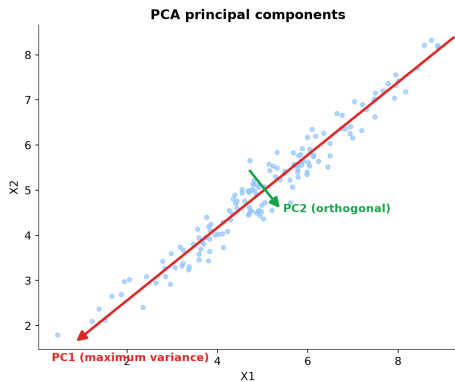
PCA: the direction of maximum variance

- PCA projects the data onto the axes where it **varies most**, losing as little information as possible
- PC1: the direction of maximum variance; PC2: orthogonal to PC1, captures the remaining variance



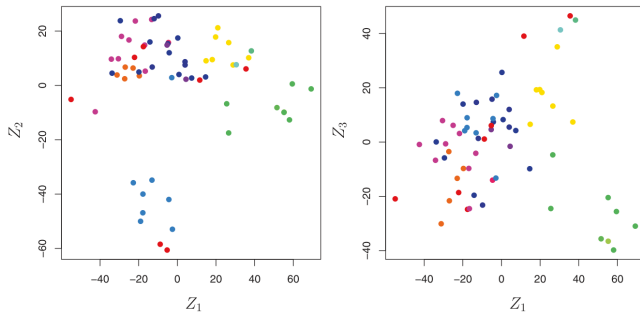
PCA: principal components

- Computed via singular value decomposition (SVD) of the data matrix
- Can be seen as a **linear autoencoder**: compress then reconstruct, minimizing the error



PCA on real data: NCI60 (6830 genes)

From 6830 genes down to just 3 principal components: lines of the same cancer type (same color) cluster together despite the huge original dimensionality.

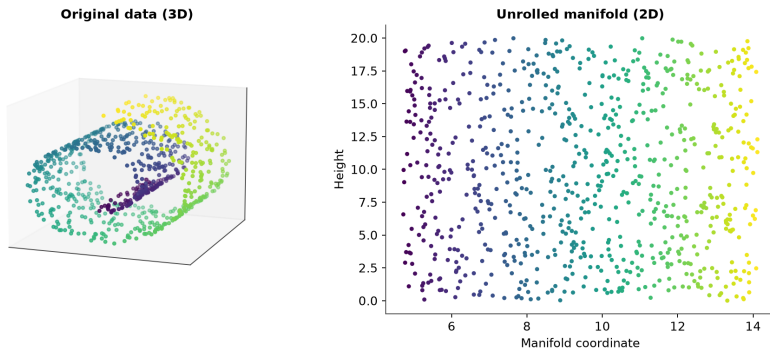


Source: James, Witten, Hastie & Tibshirani (2013), *An Introduction to Statistical Learning*, Fig. 10.15.

Manifold learning: when data isn't flat

- PCA assumes the interesting structure is **linear** — it fails on curved structures
- The **manifold hypothesis**: real-world data (images, voices...) lives on a low-dimensional curved surface embedded in a high-dimensional space

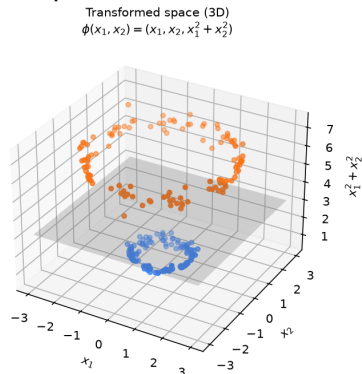
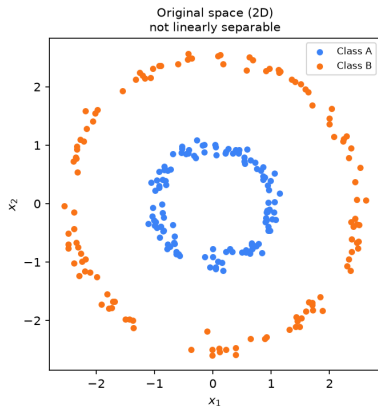
Swiss roll: the motivation for manifold learning



Kernel PCA: the kernel trick

- Maps the data into a higher-dimensional space where it *is* linearly separable
- Useful when the original data can't be split by a straight line

The kernel trick: separating the inseparable

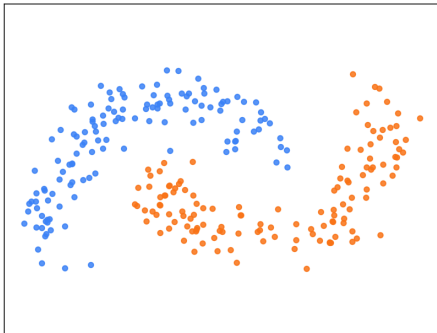


t-SNE: preserving local neighborhoods

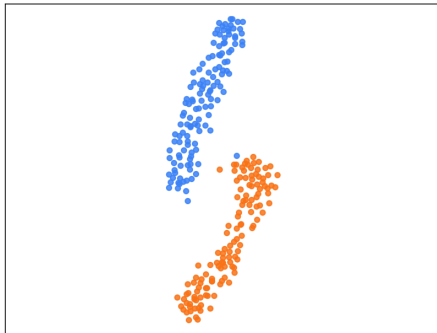
- Turns distances into neighborhood probabilities: nearby points should stay close after projection
- Excellent for **visualizing** complex clusters in 2D; slower on large datasets

PCA vs. t-SNE: two interleaved moons embedded in 30D

PCA (linear projection)
fails to separate the classes

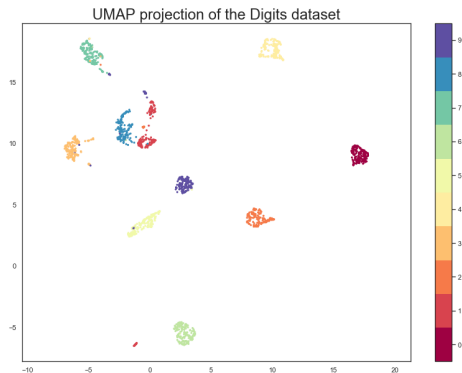


t-SNE (non-linear projection)
separates each moon into a group



UMAP: fast and accurate

- Preserves both **local** and **global** structure, unlike t-SNE
- Much faster: suitable for datasets with millions of samples



Source: *umap-learn docs*, Copyright (c) 2017 Leland McInnes, BSD-3 license.

Summary: 3.2 Dimensionality reduction

- **PCA**: fast, linear, interpretable
- **Kernel PCA**: extends PCA to non-linear structures
- **t-SNE**: excellent visualization, non-linear, slow on large data
- **UMAP**: the best balance of speed and quality

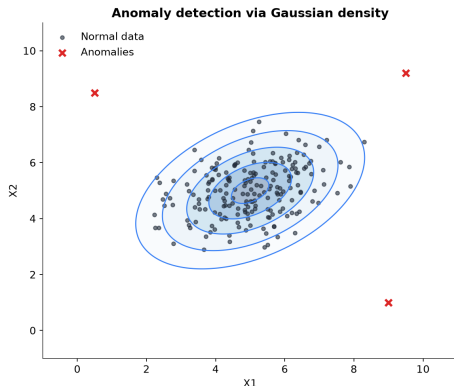
Philosophy of anomaly detection

- *Inliers* (normal): frequent, with consistent patterns
- *Outliers* (anomalies): rare, deviate significantly from the norm
- **Anomaly detection**: training data may contain noise
- **Novelty detection**: training data is completely clean

Example: bank fraud, failures on a production line, or a sensor giving strange readings: in none of these cases do we know in advance what the next anomaly will look like.

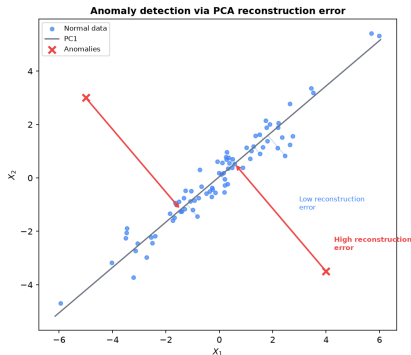
Gaussian approach

- A Gaussian distribution is fit to the normal data
- Points with **low probability density** are flagged as anomalies
- The threshold can be set from the known historical rate of rare cases



PCA reconstruction

- Data is projected to a low dimension with PCA and then reconstructed back
- Anomalies don't follow the main structure, so their **reconstruction error** is much larger



Isolation Forest

- Instead of modeling what's normal, it tries to **isolate** each point using random partitions
- Anomalies, being far from the rest, get isolated with very few partitions

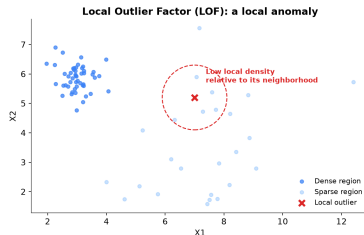
Isolation Forest: the isolation intuition



Local Outlier Factor (LOF)

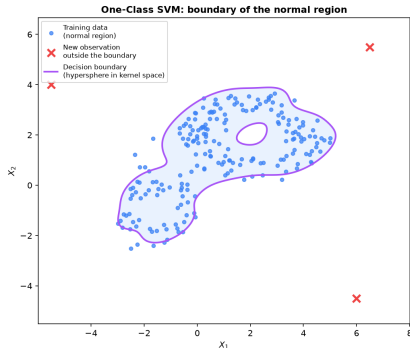
- Compares a point's **local density** with that of its neighbors
- Detects **local anomalies**: points that aren't extreme in absolute terms, but are unusual in their neighborhood

Example: a 200 m² house isn't unusual nationwide, but it is in a neighborhood where every house is 60 m²: that's exactly what LOF detects.



One-class SVM

- Designed for novelty detection: trained on clean data
- Finds the **minimum-volume** region that encloses nearly all the training data
- Any new observation falling outside that boundary is flagged as an anomaly



Summary: 3.3 Anomaly detection

- Anomalies can be rare, contextual, or collective events
- **Simple methods:** Gaussian, PCA reconstruction
- **Advanced methods:** Isolation Forest, LOF, one-class SVM

1. Foundations

- History of ML
- Overfitting/underfitting
- Cross-validation

2. Supervised

- Regression, regularization
- Classification, SVM
- Neural networks

3. Unsupervised

- Clustering
- PCA, manifold learning
- Anomaly detection

Thank you!

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